RAJALAKSHMI ENGINEERING COLLEGE

Shreya Kidangappilly 250701704-2025-CSE

S2

| | |
|---|---|
| **Started on** | Friday, 11 September 2026, 9:06 AM |
| **State** | Finished |
| **Completed on** | Friday, 11 September 2026, 9:10 AM |
| **Time taken** | 3 mins 57 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

**For example:**

| Input | Result |
|---|---|
| 5<br>1 1 2 3 4 | 1 |

**Answer:** (penalty regime: 0 %)

```c
#include <stdio.h>
int main(){
    int n,a[100],i,j;
    scanf("%d", &n);
    for(i=0;i<n;i++){
        scanf("%d", &a[i]);
    }
    for(i=0; i<n; i++){
        for(j=i+1; j<n; j++){
            if(a[i] == a[j]){
                printf("%d", a[i]);
                return 0;
            }
        }
    }
    return 0;
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 11<br>10 9 7 6 5 1 2 3 8 4 7 | 7 | 7 | ✔ |
| ✔ | 5<br>1 2 3 4 4 | 4 | 4 | ✔ |
| ✔ | 5<br>1 1 2 3 4 | 1 | 1 | ✔ |

Passed all tests! ✔

**Correct**

Marks for this submission: 1.00/1.00.

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S2

| | |
|---|---|
| **Started on** | Friday, 11 September 2026, 9:11 AM |
| **State** | Finished |
| **Completed on** | Friday, 11 September 2026, 9:17 AM |
| **Time taken** | 6 mins 14 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

**For example:**

| Input | Result |
|---|---|
| 5<br>1 1 2 3 4 | 1 |

**Answer:** (penalty regime: 0 %)

```c
#include <stdio.h>
int main(){
    int n,i,j;
    int a[100];
    scanf("%d", &n);
    for(i=0; i<n; i++){
        scanf("%d", &a[i]);
    }
    for(i=0;i<n;i++){
        for(j=i+1;j<n;j++){
            if(a[i] == a[j]){
                printf("%d", a[i]);
                return 0;
            }
        }
    }
    return 0;
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 11<br>10 9 7 6 5 1 2 3 8 4 7 | 7 | 7 | ✔ |
| ✔ | 5<br>1 2 3 4 4 | 4 | 4 | ✔ |
| ✔ | 5<br>1 1 2 3 4 | 1 | 1 | ✔ |

Passed all tests! ✔

**Correct**

Marks for this submission: 1.00/1.00.

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S2

| | |
|---|---|
| **Started on** | Friday, 11 September 2026, 9:18 AM |
| **State** | Finished |
| **Completed on** | Friday, 25 September 2026, 9:00 AM |
| **Time taken** | 13 days 23 hours |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

· The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array

2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

**For example:**

| Input | Result |
|---|---|
| 1<br>3 10 17 57<br>6<br>2 7 10 15 57 246 | 10 57 |

**Answer:** (penalty regime: 0 %)

```c
#include <stdio.h>
int main(){
    int t;
    scanf("%d", &t);
    while(t--){
        int n1, n2, i, j;
        scanf("%d", &n1);
        int a[n1];
        for(int i=0; i<n1; i++){
            scanf("%d", &a[i]);
        }
        scanf("%d", &n2);
        int b[n2];
        for(int i=0; i<n2; i++){
            scanf("%d", &b[i]);
        }
        i = 0;
        j = 0;
        while (i < n1 && j < n2) {

            if (a[i] == b[j]) {
                printf("%d ", a[i]);
```

```
            printf("%d ", a[i]);
            i++;
            j++;
        }
        else if (a[i] < b[j]) {
            i++;
        }
        else {
            j++;
        }
    }
    printf("\n");
}
return 0;
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 1<br>3 10 17 57<br>6<br>2 7 10 15 57 246 | 10 57 | 10 57 | ✔ |
| ✔ | 1<br>6 1 2 3 4 5 6<br>2<br>1 6 | 1 6 | 1 6 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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S2

| | |
|---|---|
| **Started on** | Friday, 25 September 2026, 8:42 AM |
| **State** | Finished |
| **Completed on** | Friday, 25 September 2026, 9:04 AM |
| **Time taken** | 21 mins 55 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

· The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array

2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

**For example:**

| Input | Result |
|---|---|
| 1<br>3 10 17 57<br>6<br>2 7 10 15 57 246 | 10 57 |

**Answer:** (penalty regime: 0 %)

```
#include <stdio.h>
int main(){
    int t;
    scanf("%d", &t);
    while(t--){
    int n1, n2, i, j;
    scanf("%d", &n1);
    int a[n1];
    for(int i=0; i<n1; i++){
        scanf("%d", &a[i]);
    }
    scanf("%d", &n2);
    int b[n2];
    for(int i=0; i<n2; i++){
        scanf("%d", &b[i]);
    }
    i=0;
    j=0;
    while (i < n1 && j < n2)
        {
            if (a[i] == b[j])
            {
```

```
            {
                printf("%d ", a[i]);
                i++;
                j++;
            }
            else if (a[i] < b[j])
            {
                i++;
            }
            else
            {
                j++;
            }
        }
        printf("\n");

    }
    return 0;
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 1<br>3 10 17 57<br>6<br>2 7 10 15 57 246 | 10 57 | 10 57 | ✔ |
| ✔ | 1<br>6 1 2 3 4 5 6<br>2<br>1 6 | 1 6 | 1 6 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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S2

| | |
|---|---|
| **Started on** | Friday, 25 September 2026, 9:18 AM |
| **State** | Finished |
| **Completed on** | Friday, 25 September 2026, 9:29 AM |
| **Time taken** | 10 mins 57 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that A[j] - A[i] = k, i != j.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as 5 - 1 = 4

So Return 1.

**For example:**

| Input | Result |
|---|---|
| 3<br>1 3 5<br>4 | 1 |

**Answer:** (penalty regime: 0 %)

```c
#include <stdio.h>
int main(){
    int n,k;
    scanf("%d", &n);
    int a[n];
    for(int i=0; i<n; i++){
        scanf("%d", &a[i]);
    }
    scanf("%d", &k);
    int i=0, j=1;
    while(i<n && j<n){
        if(i==j){
            j++;
        }
        else{
            int diff = a[j]-a[i];
            if(diff == k){
                printf("1\n");
                return 0;
            }
            else if(diff<k){
                j++;
            }
            else{
                i++;
            }
        }
    }
    printf("0\n");
    return 0;
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 3<br>1 3 5<br>4 | 1 | 1 | ✔ |
| ✔ | 10<br>1 4 6 8 12 14 15 20 21 25<br>1 | 1 | 1 | ✔ |
| ✔ | 10<br>1 2 3 5 11 14 16 24 28 29<br>0 | 0 | 0 | ✔ |
| ✔ | 10<br>0 2 3 7 13 14 15 20 24 25<br>10 | 1 | 1 | ✔ |

Passed all tests! ✔

**Correct**

Marks for this submission: 1.00/1.00.

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Shreya Kidangappilly 250701704-2025-CSE

S2

| | |
|---|---|
| **Started on** | Friday, 25 September 2026, 8:49 AM |
| **State** | Finished |
| **Completed on** | Friday, 25 September 2026, 9:16 AM |
| **Time taken** | 26 mins 33 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that A[j] - A[i] = k, i != j.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as 5 - 1 = 4

So Return 1.

**For example:**

| Input | Result |
|---|---|
| 3<br>1 3 5<br>4 | 1 |

**Answer:** (penalty regime: 0 %)

```c
#include <stdio.h>
int main(){
    int n, k;
    scanf("%d", &n);
    int a[n];
    for(int i=0; i<n; i++){
        scanf("%d", &a[i]);
    }
    scanf("%d", &k);
    int i=0, j=1;
    while(i<n && j<n){
        if(i==j){
            j++;
        }
        else{
            int diff = a[j]-a[i];
            if(diff == k){
                printf("1\n");
                return 0;
            }
            else if(diff<k){
                j++;
            }
            else{
                i++;
            }
        }
    }
    printf("0\n");
    return 0;
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 3<br>1 3 5<br>4 | 1 | 1 | ✔ |
| ✔ | 10<br>1 4 6 8 12 14 15 20 21 25<br>1 | 1 | 1 | ✔ |
| ✔ | 10<br>1 2 3 5 11 14 16 24 28 29<br>0 | 0 | 0 | ✔ |
| ✔ | 10<br>0 2 3 7 13 14 15 20 24 25<br>10 | 1 | 1 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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